

Mohd Arshad

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SUMMARY

Backend Engineer specializing in **Distributed Systems, Event-Driven Architecture, and High-Throughput Microservices**. Core expertise in **Java, Spring Boot, Apache Kafka, and Redis**, with hands-on focus on solving **distributed concurrency, database consistency, and fault tolerance**. Experienced in container orchestration and integrating AI-driven pipelines into production systems.

EDUCATION

Babu Banarsi Das Northern India Institute Of Technology

Lucknow, India

Bachelor of Technology in Information Technology (GPA: 7.87/10.0)

2022 – 2026

TECHNICAL SKILLS

Languages: Java, Python, SQL (PostgreSQL), TypeScript

Backend & Distributed Systems: Spring Boot 3, Apache Kafka, Redis (Redlock/Caching), Microservices, REST APIs, Resilience4j

Databases & Storage: PostgreSQL, MongoDB, Redis, HikariCP, Indexing & Query Optimization

Cloud, DevOps & Observability: Docker, Kubernetes, Prometheus, Grafana, Distributed Tracing (Zipkin/OpenTelemetry), Git/GitHub

AI Engineering & Tools: Spring AI, Gemini API, Vector DBs (pgvector), Testcontainers, Postman, Maven

EXPERIENCE

Software Engineer Intern

Mar 2025 – Aug 2026

HireEase (AplyEase)

Remote

- Engineered an autonomous multi-agent AI pipeline utilizing **Google Gemini API** to perform contextual extraction and iterative resume optimization, elevating ATS candidate-match scores to **90%+**.
- Spearheaded backend architecture and cross-functional engineering workflows as interim lead, standardizing structured JSON prompt schemas and modular delivery pipelines for an agile team.
- Engineered automated document generation workflows and asynchronous worker handlers, cutting end-to-end client delivery turnaround time by **40%** across high-volume production requests.

RESEARCH & PUBLICATIONS

GeoSentinel: Computational Framework for Conflict Simulation

Nov 2025 – May 2026

- Engineered a synthetic simulation framework in **Python** to model non-linear dynamic interactions between physical conflict events and narrative sentiment across discrete state transitions.
- Developed a dual-pillar NLP pipeline integrating fine-tuned **DistilBERT** for high-density sentiment quantification alongside **PCA** for dynamic feature extraction and dimensionality reduction.
- Applied **Granger Causality tests** on multidimensional time-series data to statistically validate predictive lead-lag relationships between narrative spikes and physical escalation events.

PROJECTS

SentinelLedger | *Java 21, Spring Boot 3, Kafka, Redis, PostgreSQL, ONNX Runtime, Docker*

- Architected an event-driven payment engine processing **10,000+ RPS**, enforcing strict append-only double-entry bookkeeping schemas to eliminate balance tampering and ensure immutable auditability.
- Implemented the **Saga Orchestration Pattern** with a **Transactional Outbox** to guarantee at-least-once message delivery and automatic compensating rollbacks across distributed microservices.
- Eliminated double-spending vulnerabilities and race conditions under peak concurrency using **Redis Redlock** paired with cryptographic idempotency validation keys.
- Integrated an embedded **XGBoost fraud model via Java ONNX Runtime**, achieving sub-**15ms** inference latency by executing in-memory feature scoring and bypassing remote network hops.

FlashTix | *Java 21, Spring Boot 3, Redis, PostgreSQL, Lua Scripts, Prometheus, Grafana*

- Engineered a distributed ticketing platform handling **5,000+ RPS** without overselling anomalies, utilizing atomic **Redis Lua scripts** for rapid in-memory seat allocation and state management.
- Enforced multi-layered data consistency by combining **Redis Distributed Locking** with **PostgreSQL Optimistic Locking (@Version)** as a resilient fail-safe boundary under heavy database contention.
- Tuned database connection pooling via **HikariCP** to maximize throughput and established live telemetry monitoring for lock contention, error rates, and p99 latency using **Prometheus and Grafana**.

StreamFlow | *Java, Spring Boot, Apache Kafka, Redis, MongoDB, Zipkin, Docker*

- Architected a scalable messaging backbone leveraging **Apache Kafka** custom partition key hashing and manual offset management, guaranteeing at-least-once delivery semantics across worker pools.
- Configured an automated fault-recovery pipeline utilizing **Dead Letter Queues (DLQ)** with exponential backoff and jitter algorithms to isolate and retry transient downstream provider failures.
- Implemented a **Redis Write-Through Caching** layer to serve high-frequency notification reads at sub-**5ms** latency, offloading **70%+** read IOPS from the underlying **MongoDB** database.